



ON-BOARD WITH MOTOTRBO™ LINKED CAPACITY PLUS

BUS COMPANY DEBUTS FIRST DIGITAL TRUNKED RADIO SOLUTION FOR ENHANCED FLEET SERVICES



Stagecoach Bus Manchester is the first company worldwide to deploy the MOTOTRBO Linked Capacity Plus system to improve its operational communications. The digital two-way radio system makes it possible to communicate throughout a wide area by linking adjoining sites across an IP network.

Part of a large bus operator in the United Kingdom, Stagecoach Group, the company operates nearly 700 buses across the greater Manchester area and relies on reliable, ubiquitous radio coverage to maintain communication with four control centres.

The MOTOTRBO Linked Capacity Plus system was supplied and implemented by Motorola Solutions' local partner, Affini, formerly AirRadio. It provides essential features, including: call dispatching and logging, group calls, telephone interconnect to allow calls to be made from the MOTOTRBO radios and any fixed line or mobile telephone across the operation, and Transmit Interrupt (to override a call) for the instant delivery of emergency communications.

CUSTOMER PROFILE

Company

Stagecoach UK Bus
(Manchester Operation)

Partner Name:

Affini – formerly AirRadio

Industry Name:

Transportation

Product Name:

- MOTOTRBO Linked Capacity Plus
- DP 3600 portable & DM 3400 mobile radios
- DR 3000 repeater
- TRBOnet Enterprise 3.5 Dispatcher Application

Key Benefits:

- Scalable, seamless, multi-site communications
- Improved safety and security features
- Voice and data integration with enhanced voice quality
- Increased productivity and efficiency

CASE STUDY

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“Affini came with the vision, the backing of Motorola Solutions and years of expertise that made the decision to choose Linked Capacity Plus easy. We could see immediate infrastructure cost savings by moving from the existing five site MPT1327 solution to a two site digital platform whilst maintaining functionality and radio coverage. The possibility of this model being deployable to other areas of our business was key to gaining the executive team’s approval.”

Peter Sumner, Engineering Director at Stagecoach Manchester



THE CHALLENGE

Stagecoach had been using an MPT1327 trunked radio network which was shared with Manchester’s second bus company, First Bus Manchester. But as it had been in operation for over twenty years, the cost of operating and maintaining it had become very high. In addition there was a pressing regulatory requirement to change the operating frequencies which meant a costly replacement of most of the RF system hardware.

Stagecoach approached trunked radio specialist and Motorola Solutions partner, Affini during 2011 with a view to migrating to digital radio technology and MOTOTRBO Linked Capacity Plus, a multi-site digital trunked radio solution, was recommended.

Affini planned, designed and implemented the Linked Capacity Plus system to meet the specific needs of Stagecoach Manchester’s operation. This included the provision of four control centres that share all the radio channels and are responsible for co-ordinating the movement of over 690 city buses.

THE SOLUTION

The Linked Capacity Plus implementation began in early 2012 and involved using two existing sites for the deployment of new repeater and antenna equipment. The main site is located in Manchester’s city centre and the second site is on the 280m high Werneth Low hill on the outskirts of the city to provide ‘fill-in signal’ for total coverage. The system comprises around 700 DM 3400 mobile radios installed in the buses, 50 DP 3600 portable radios for operations personnel and TRBOnet Enterprise 3.5 fixed PC Dispatcher Stations deployed at each of the four operations centres.

So how does the system work? A bus driver presses an overhead button to request a call-back and a dispatcher at one of the operations centres responds to find out what the request entails. The driver will then use a foot pedal to communicate the request and the operations centre responds and provides information on how to handle the situation, for example, a traffic jam.

Should the operations centre need to share any information with a larger, defined, group of buses they can make a broadcast call. In an emergency the bus driver can use the foot pedal to over-ride busy channels and get through to the dispatcher instantly. Requests may include information about lost property or the police can request information when tracing the movements of people who are known to have used the buses.

THE BENEFIT

Linked Capacity Plus was designed to provide seamless, multisite, scalable and easy-to-use communications to organisations, such as Stagecoach, that have large operations spread over a wide area. With seamless roaming, drivers can focus on the task at hand without having to adjust their radios as they travel throughout the coverage area – ensuring the safety and security of both drivers and passengers. The foot pedal enables hands-free communications which is essential for drivers.

Voice and data integration in a single system eliminates the need for separate network controllers and allows the company to be more efficient by using key applications, such as the TRBOnet Enterprise Dispatcher to respond quickly to driver requests and direct bus crews. The inclusion of telephone interconnect provides seamless communication between radios and landline or mobile phones.

Linked Capacity Plus supports a safer work environment through key features such as Transmit Interrupt to prioritise important communication exactly when it’s needed and emergency alerts which enable drivers to send notifications to a central location instantly.

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